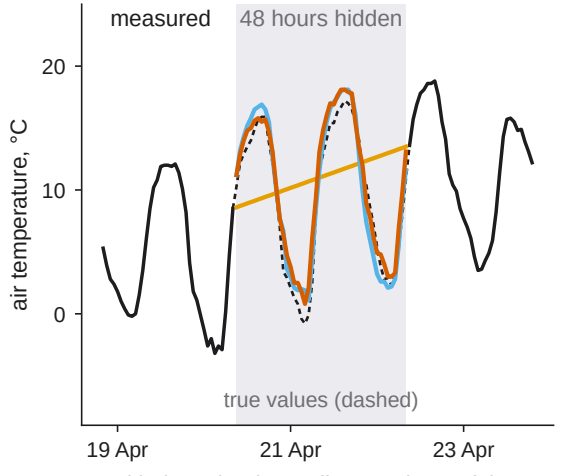
1 Hide known hours



Stockholm-Arlanda, April 2023: three of the seven methods filling one real gap

2 Refill and simulate

- Linear interpolation
- Daily patterns
- Four-week average
- Nearby station
- ERA5 reanalysis
- **Stepwise, with and without ERA5 switch**

seven methods

17

weather-years,
6 Swedish regions

3 h to 1 week

gaps, at random or in heatwaves and cold spells

6

buildings, EnergyPlus

each filled file is compared with the complete one

3 What changes

< 0.3% change in annual heating with 12-hour gaps, in 95% of cases

½ of heating peaks hidden by a week-long gap move by more than a day

1–2 days then switch to ERA5: this avoids the worst errors, from nearby stations failing in extreme weeks

Quick check: extra heating ≈ heat-loss coefficient × temperature error summed over the gap